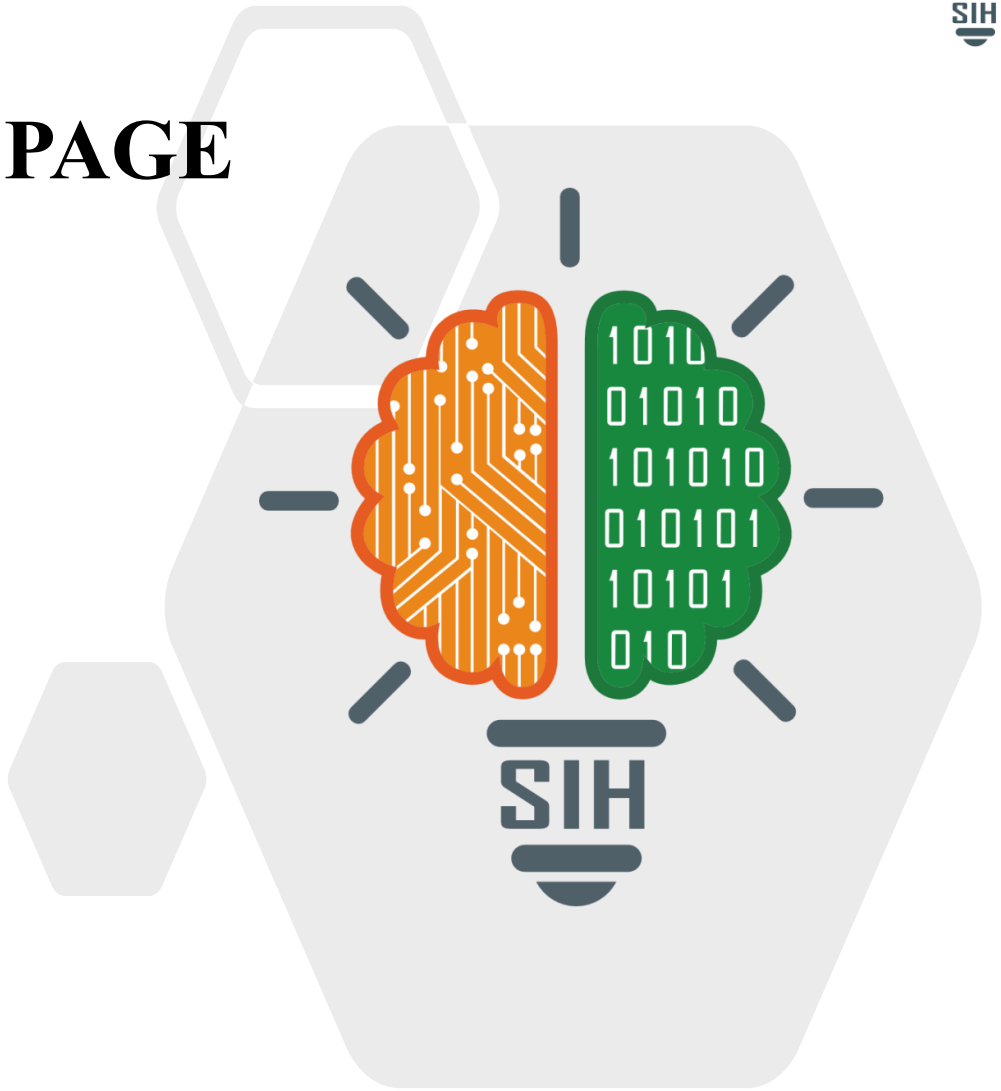
TITLE PAGE

- **Problem Statement ID –** 26172
- **Problem Statement Title –** Low Latency and Efficient Voice Activator for Edge Devices
- **Theme –** Smart Automation
- **PS Category –** Hardware
- **Team ID –** AUC-SIH26-172041
- **Team Name –** DATA_Edgepulse



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Hey Vayro: Edge Wake Word, Cloud ASR

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❖ Proposed Solution (Describe your Idea/Solution/Prototype)

Detailed explanation of the proposed solution

- ESP32-S3 + INMP441 microphone listen continuously; every 10 ms the chip computes a 40-band log-mel frame
- A 10.7 k-parameter DS-CNN (28 KB int8, TensorFlow Lite Micro + ESP-NN) scores the last 1 s every 60 ms
- “Hey Vayro” detected (mean of 3 scores ≥ 0.80) → 571 ms pre-roll + live audio stream over one warm TCP link
- Server: VAD ends the utterance, Vosk (Kaldi) ASR returns the command text

How it addresses the problem

- Audio leaves the device only after a detection (true or false) → privacy; nothing is sent or processed while idle
- Inside the PS budget on real hardware: 245.0 KiB peak RAM, 5.5 % idle CPU
- Low latency: keyword end → audio at the ASR server 138 ms median (196 ms p95), first audio 18 ms after detection
- No proprietary voice SDK: keyword pipeline fully open source (only the Wi-Fi radio driver is closed, disclosed); keyword trained from scratch

Innovation and uniqueness of the solution

- Training and device share one bit-exact feature pipeline (12 000 / 12 000 frames identical)
- Pre-roll: the stream starts 571 ms before the detection, so a command spoken right after the keyword is not cut off
- The device logs every trigger with its score, so real-home false triggers can be mined for retraining
- Selection rules written before each held-out test (one documented override); 63 evidence entries

28 KB
int8 model · 13.4 KB tensor arena

245.0 KiB
peak RAM while streaming (< 256 KB)

5.5 %
idle CPU, both cores (busiest core 9.5 %)

138 ms
keyword end → ASR server (median)

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TECHNICAL APPROACH

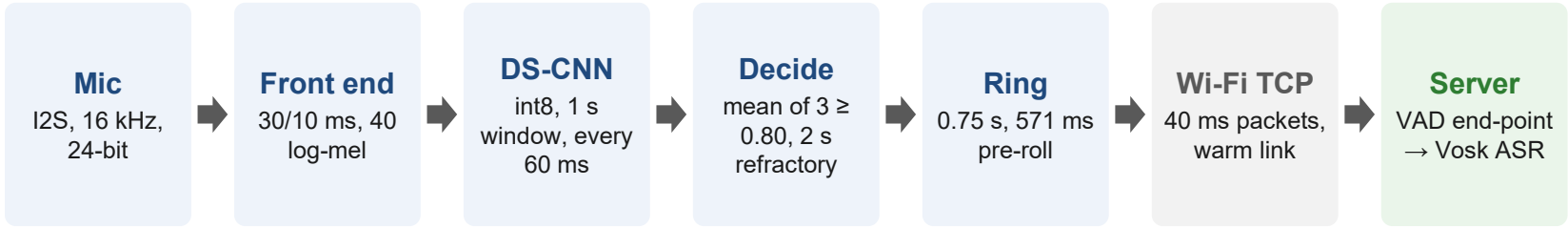
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❖ Technologies to be used (e.g. programming languages, frameworks, hardware)

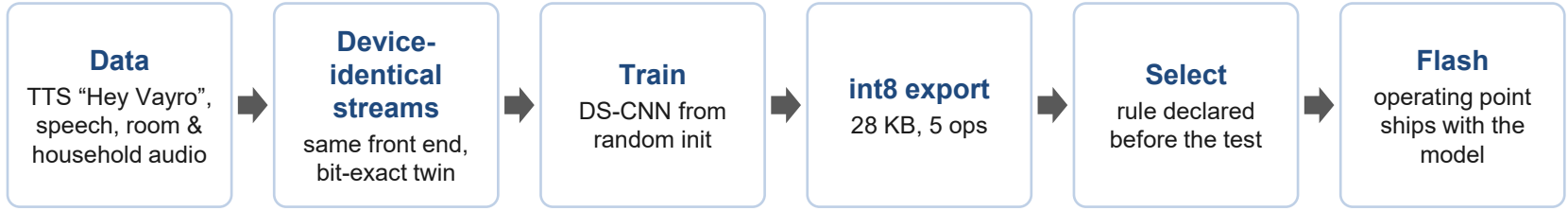
- **Hardware:** ESP32-S3 N16R8 — 240 MHz dual core, SIMD used by ESP-NN, on-chip Wi-Fi (PSRAM off); INMP441 — digital I2S mic, no analog stage
- **Firmware:** C / C++ on ESP-IDF 5.5.5 (FreeRTOS) — open source, vendor-supported
- **TinyML:** TensorFlow Lite Micro + ESP-NN, TFLM microfrontend — integer inference, no OS needed
- **Model:** DS-CNN — small and accurate for keyword spotting on MCUs [1]
- **Training:** Python, TensorFlow / Keras, Piper TTS (4 voices), LibriSpeech, our own recordings
- **Server:** Python ingest, energy VAD, Vosk / Kaldi — open, offline, not a transformer
- **Verification:** bit-exact host twin of the front end, scripted capture and replay

❖ Methodology and process for implementation (Flow Charts/Images/ working prototype)

Runtime — on the ESP32-S3 until the keyword, then the server



How the keyword model is made (all scripted, reproducible)



Working prototype: ESP32-S3-N16R8 + INMP441 (SCK→GPIO6, WS→GPIO5, SD→GPIO4). The demo video shows keyword → stream → transcript live on this board.

Model: 10 665 parameters · 212 584 MACs per inference · 28 024 B int8 · tensor arena 13 404 B · ops: Conv2D, DepthwiseConv2D, FullyConnected, Reshape, Logistic

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FEASIBILITY AND VIABILITY

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❖ Analysis of the feasibility of the idea — measured on the physical prototype, final firmware

Efficiency

245.0 KiB peak RAM, streaming (< 256 KB)	**5.5 %** idle CPU, 2 cores (max core 9.5 %)	**28 KB** model flash · 13.4 KB arena

Accuracy (shipped model v2b)

22 / 25 speaker 1, held-out session, 0.5–2 m	**20 / 25** new speaker, never trained on	**0.2 · 15.8** false triggers per hour: quiet · TV/talk

Latency (final firmware, ASR on)

138 ms keyword end → ASR server (median)	**196 ms** keyword end → ASR server (p95)	**18 ms** detection → first audio (median)

❖ Potential challenges and risks	❖ Strategies for overcoming these challenges
TV / family speech still triggers: 15.8 per hour (2.5 h of held-out household audio)	On the same 52 min, retraining with the device's own false triggers cut 15.1 → 10.4 /h (v4), but v4 lost new-speaker recall (14/25), so v2b ships; next: more speakers + household audio together
Real voices tested so far: two speakers (plus 4 TTS voices: US, UK, non-native, Hindi)	Scripted 3-minute sessions for new speakers; per-speaker results reported separately
Server ASR word accuracy is set by the ASR model (Vosk en-in: 36 % WER published)	Edge is ASR-agnostic: swap in a larger model or a command grammar on the server only
Thin RAM margin; Wi-Fi or server outages	Memory trims saved 9.6 KB; detection keeps running offline; auto-reconnect; ASR off the server's receive path

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IMPACT AND BENEFITS

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❖ Potential impact on the target audience

- **ISRO labs & control rooms:** hands-free voice commands where hands and eyes are busy — without streaming room audio all day
- **Smart automation:** a ready wake-word front end for ESP32-S3 home, office or factory devices
- **Assistive use:** voice control for elderly or motor-impaired users on low-cost hardware
- **Indian developers & start-ups:** custom keyword with Indian-accent training data and no vendor voice SDK

0 bytes
of audio sent before a detection

32 kB/s
16 kHz audio, only from keyword to end of command

❖ Benefits of the solution (social, economic, environmental, etc.)

Social

Privacy by design: audio leaves the chip only after a detection (true or false); nothing is sent while idle

Economic

Hardware ≤ ₹2,000 (team estimate); open-source stack; no audio processed in the cloud while idle

Environmental

Low compute (5.5 % CPU idle); audio is sent only after a detection, so traffic stays tiny. Wi-Fi is kept awake for latency; power not yet measured

Technical

Reproducible: data, training, selection and tests are scripts; every number is traceable to an evidence log

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RESEARCH AND REFERENCES

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❖ Details / Links of the reference and research work

[1] **Y. Zhang, N. Suda, L. Lai, V. Chandra,** “Hello Edge: Keyword Spotting on Microcontrollers”, arXiv:1711.07128, 2017 — DS-CNN architecture

[2] **P. Warden,** “Speech Commands: A Dataset for Limited-Vocabulary Speech Recognition”, arXiv:1804.03209, 2018 — KWS background (not used for training)

[3] **TensorFlow Lite Micro + microfrontend:** github.com/tensorflow/tflite-micro

[4] **Espressif esp-tflite-micro and ESP-NN:** github.com/espressif/esp-tflite-micro · github.com/espressif/esp-nn

[5] **ESP-IDF v5.5 (FreeRTOS, drivers, lwIP):** github.com/espressif/esp-idf

[6] **V. Panayotov et al.,** “LibriSpeech: an ASR corpus based on public domain audio books”, ICASSP 2015 — openslr.org/12

[7] **Piper TTS and voices:** github.com/OHF-Voice/piper1-gpl · huggingface.co/rhasspy/piper-voices

[8] **D. Povey et al.,** “The Kaldi Speech Recognition Toolkit”, ASRU 2011 · Vosk: alphacephei.com/vosk

[9] **Datasheets:** ESP32-S3 Technical Reference Manual (Espressif); INMP441 (TDK InvenSense)

Our own research record

- **Evidence ledger:** 63 entries, each tagged measured / estimated / proposed and pointing to its raw data
- **Selection rules:** written and committed before each held-out test; one documented override (new-speaker evidence)
- **SBOM:** every component and licence listed; the closed Wi-Fi/PHY radio driver and two non-commercial TTS voices are disclosed
- **Licence:** our code Apache-2.0; every component’s licence listed in the SBOM
- **Demo video:** link to be added